



# Towards Predictive Science for Controllable Learning Systems

## Previous Work, Curiosity and Future Vision

Keyu Wang  
September, 2026

# Background

## ➤ Education

M.Sc. of Machine Learning  
*University of Tuebingen*

Oct. 2024 – May 2027 (Expected)  
*Tuebingen, Germany*

B.Eng of Artificial Intelligence  
*Southeast University*

Sep. 2020 – Jun. 2024  
*Nanjing, China*

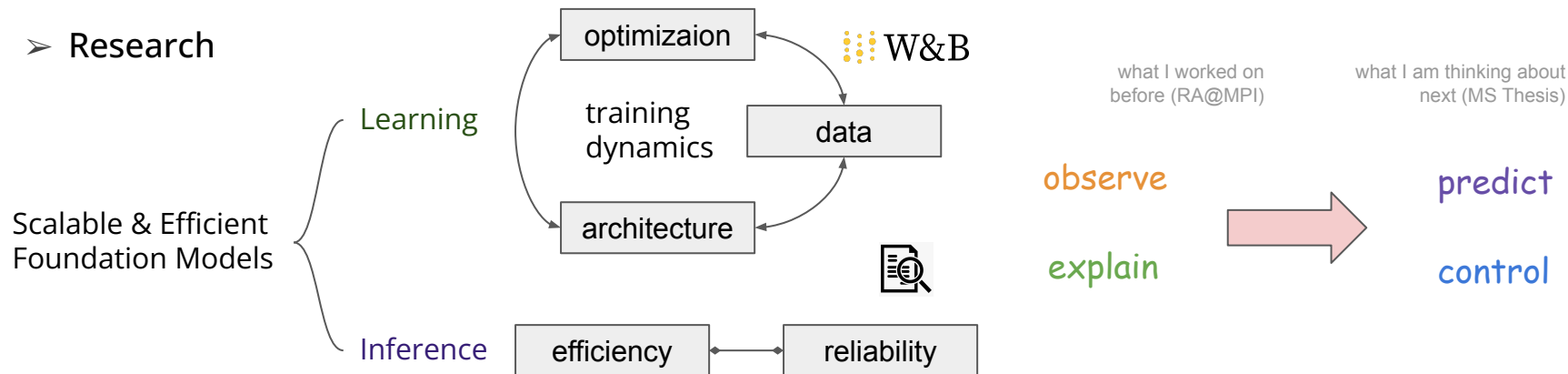
## ➤ Experience

Research Assistant  
Max Planck Institute for Intelligent Systems

Dec. 2025 – Jan. 2027 (Expected)  
*Tuebingen, Germany*

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## ➤ Research





**MAX PLANCK INSTITUTE**  
FOR INTELLIGENT SYSTEMS



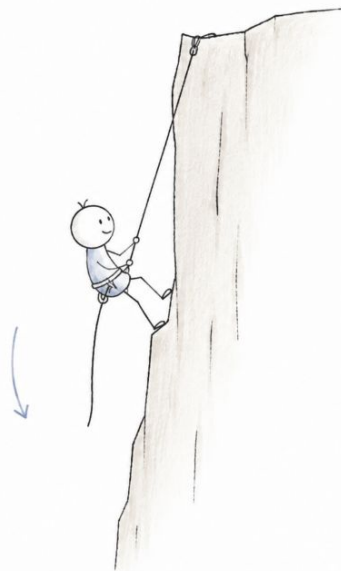
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INSTITUTE  
TÜBINGEN

# Towards Effective Depth Scaling in LLMs

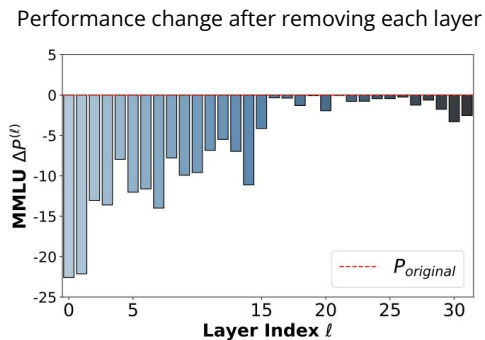
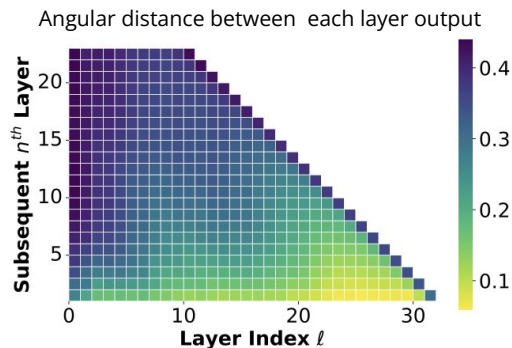
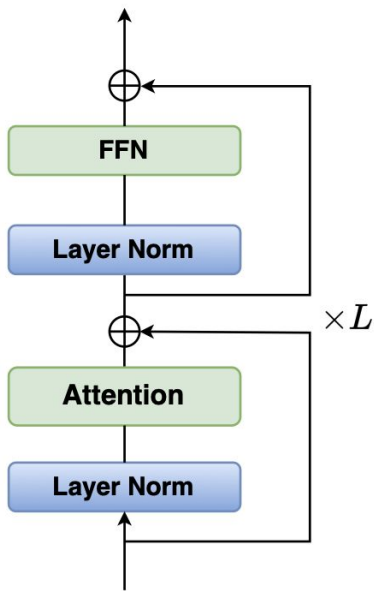
## My Recent Research Line

Keyu Wang  
Advisor: Dr. Shiwei Liu

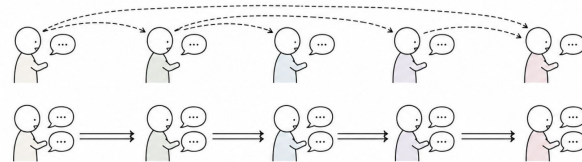
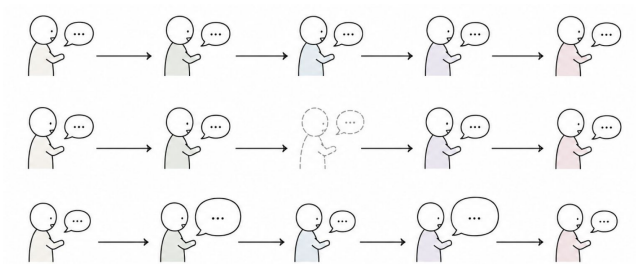
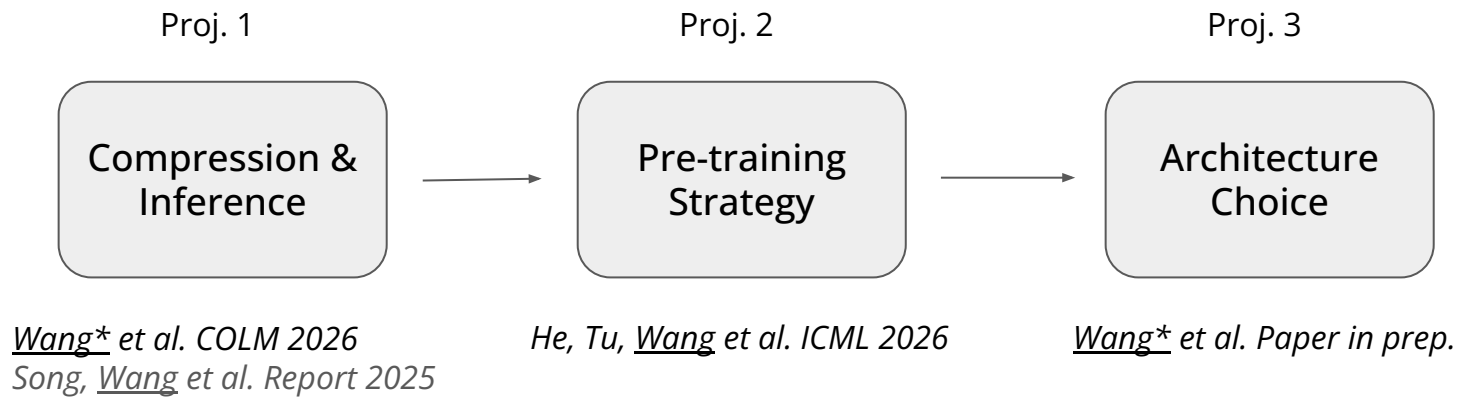


# Motivation: depth scaling pays diminishing returns

- Pre-LN dominates in modern LLMs  
— simple yet **stable** at scale
- But, Pre-LN faces the **Curse of Depth**  
— deep layers become similar and ineffective



# Outline



# Project 1: Conference on Language Modeling - 2026

## When Fewer Layers Break More Chains: Layer Pruning Harms Test-Time Scaling in LLMs

**Keyu Wang**<sup>1,2,3\*</sup> **Tian Lyu**<sup>4\*</sup> **Guinan Su**<sup>1,2,3</sup> **Lu Yin**<sup>5</sup> **Marco Canini**<sup>4</sup>  
**Jonas Geiping**<sup>1,2,3</sup> **Shiwei Liu**<sup>1,2,3</sup>

<sup>1</sup>Max Planck Institute for Intelligent Systems    <sup>2</sup>ELLIS Institute Tübingen

<sup>3</sup>Tübingen AI Center    <sup>4</sup>King Abdullah University of Science and Technology

<sup>5</sup>University of Surrey

keyu.wang@tuebingen.mpg.de    sliu@tue.ellis.eu

# Project 1: Motivation

- The **curse** of depth seems to be a **blessing** for layer pruning...

ShortGPT: Layers in Large Language Models are **More Redundant** Than You Expect

Xin Men\*  
Baichuan Inc.

Mingyu Xu\*  
Baichuan Inc.

Qingyu Zhang\*  
ISCAS

Bingning Wang†  
Baichuan Inc.

Hongyu Lin  
ISCAS

Yaojie Lu  
ISCAS

Xianpei Han  
ISCAS

Weipeng Chen  
Baichuan Inc.

PRUNE&COMP: **Free Lunch** for Layer-Pruned LLMs via Iterative Pruning with Magnitude Compensation

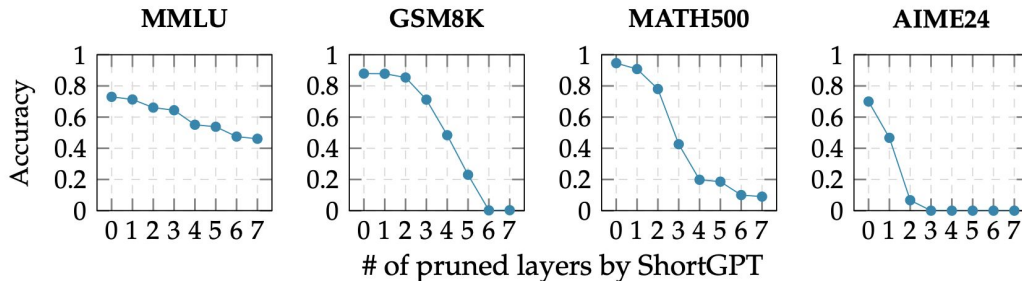
Xinrui Chen<sup>1</sup>, Hongxing Zhang<sup>2</sup>, Fanyi Zeng<sup>1</sup>, Yongxian Wei<sup>1</sup>, Yizhi Wang<sup>1</sup>,  
Xitong Ling<sup>1</sup>, Guanghao Li<sup>1</sup>, Chun Yuan<sup>1\*</sup>

<sup>1</sup> Shenzhen International Graduate School, Tsinghua University

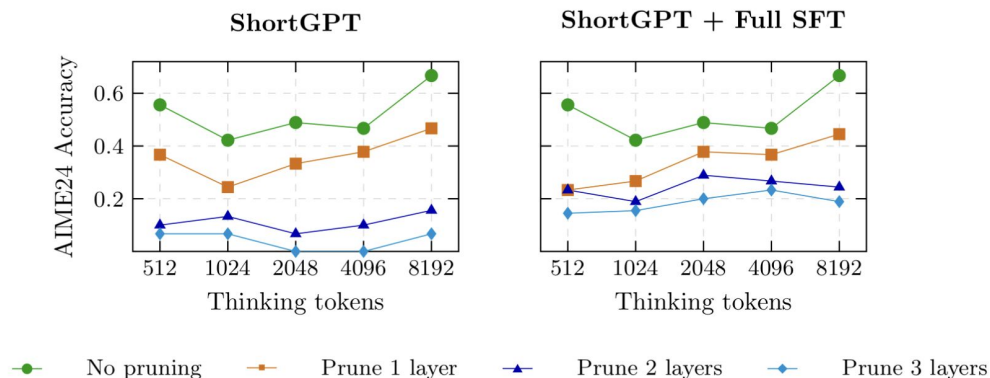
<sup>2</sup> School of Information Science and Technology, Guangdong University of Foreign Studies  
cxr22@tsinghua.org.cn, yuanc@sz.tsinghua.edu.cn

Recent studies reveal that the deeper layers (Transformer blocks) in modern LLMs tend to be less effective than the earlier ones (Gromov et al., 2024; Li et al., 2024b; Men et al., 2024; Yin et al., 2024). On the one hand, this interesting observation provides an effective indicator for LLM compression. For instance, we can compress deeper layers significantly more (Dumitru et al., 2024; Lu et al., 2024; Yin et al., 2024) to achieve high compression ratios. Even more aggressively, entire deep layers can be pruned completely without compromising performance (Muralidharan et al., 2024; Siddiqui et al., 2024).

- But layer pruning **collapses** complex reasoning!  
Were we deceived by previous benchmarks of multiple-choice questions?



# Project 1: Key Findings



- Sequential test-time scaling is highly **fragile** under layer pruning
- SFT can't lasso the runaway stallion

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## [AIME24]

Let  $p$  be the least prime number for which there exists a positive integer  $n$  such that  $n^4 + 1$  is divisible by  $p^2$ . Find the least positive integer  $m$  such that  $m^4 + 1$  is divisible by  $p^2$ . **Answer:** 110

## Before Finetuning (Pruned Only)

The smallest prime  $p$  congruent to 1 modulo 8 is 17... 17 divides 83521-17=83504. Therefore, 17 divides 83521-17, so 17 divides 83521-17. Therefore, 17 divides 83521-17. Therefore, 17 divides 83521-17...

## After Finetuning (Pruned + SFT)

Let check small primes. If  $p = 2$ ,  $n^4 + 1 \equiv 2 \pmod{4}$ . If  $p = 3$ ,  $n^4 + 1 \equiv 2 \pmod{9}$ . ... If  $p = 11$ ,  $n^4 \equiv -1 \equiv 10 \pmod{11}$ , if  $n = 5$ ,  $n^4 + 1 = 626 \equiv 10 \pmod{11}$ ... The least prime  $p$  for which there exists  $n$  such that  $p^2 \mid n^4 + 1$  is  $p = 11$ . When  $p = 11$ , we need  $n^4 \equiv 10 \pmod{11}$ . If  $n = 5$ ,  $n^4 = 625 \equiv 10 \pmod{11}$ ... Now we need  $m^4 + 1 \equiv 0 \pmod{121}$ . If  $m = 5$ ,  $5^4 + 1 = 626 \equiv 120 \pmod{121}$ . Final Answer: 5.

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# Project 2: International Conference on Machine Learning - 2026

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## **One LR Doesn't Fit All: Heavy-Tail Guided Layerwise Learning Rates for LLMs**

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**Di He**<sup>1,2,3</sup> **Songjun Tu**<sup>2,3</sup> **Keyu Wang**<sup>4</sup> **Lu Yin**<sup>5†</sup> **Shiwei Liu**<sup>6,7,8†</sup>

# Project 2: Motivation

- Transformer layers are naturally trained **heterogeneously**

ARTICLE

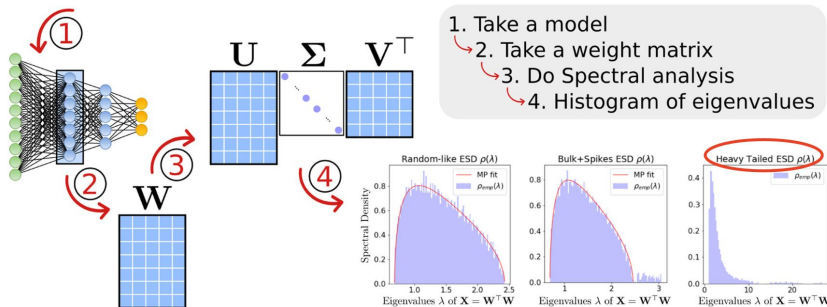
<https://doi.org/10.1038/s41467-021-24025-8>

OPEN

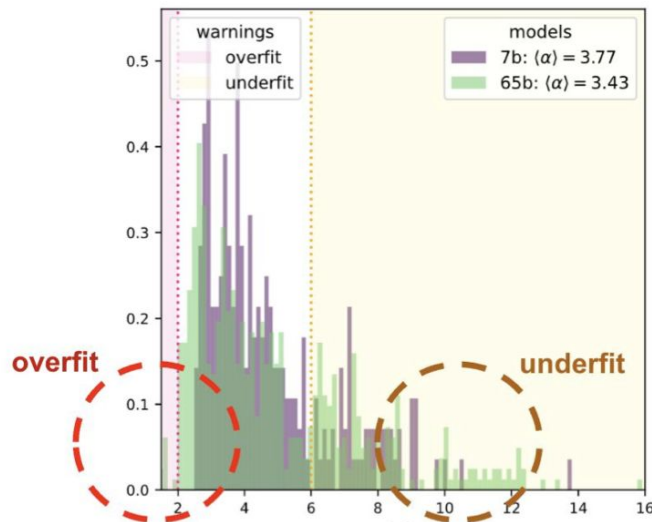


Predicting trends in the quality of state-of-the-art neural networks without access to training or testing data

Charles H. Martin<sup>1</sup>, Tongsu (Serena) Peng<sup>1</sup> & Michael W. Mahoney<sup>2</sup>



## Llama



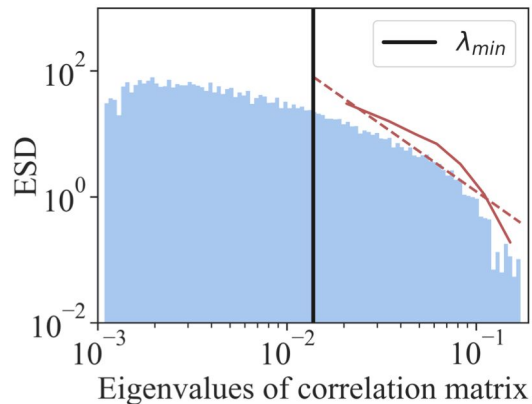
# Project 2: Algorithm & Key Results

➤ Measure **Heavy-tail** structure per-layer

$$p(\lambda) \propto \lambda^{-\alpha}$$

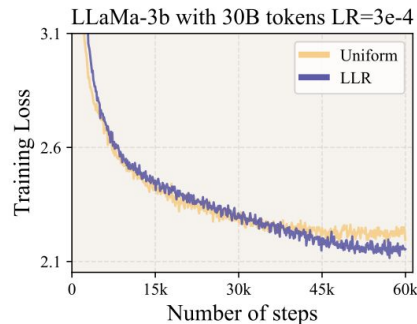
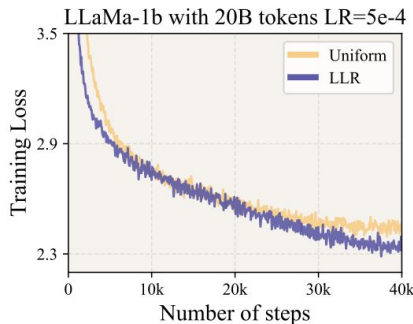
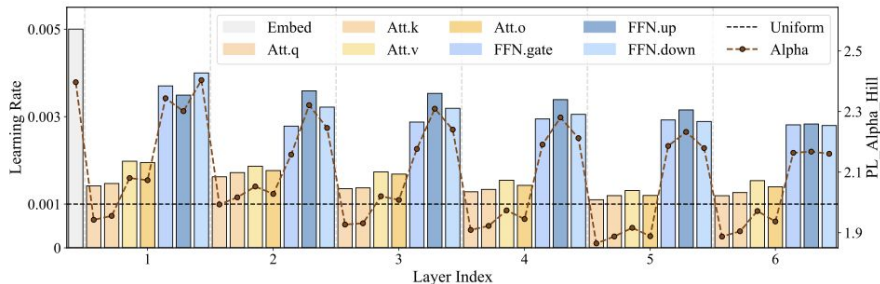
↑  
estimate

$$\text{PL\_Alpha\_Hill} = 1 + \frac{k}{\sum_{i=1}^k \ln \frac{\lambda_{n-i+1}}{\lambda_{n-k}}}$$



➤ Transfer to **layerwise LR**

$$\eta_i^T = \eta \left[ 1 + (s-1) \frac{\alpha_i^T - \alpha_{min}^T}{\alpha_{max}^T - \alpha_{min}^T} \right]$$



# Project 3: Paper is in cooking

## Architecture-dependent Aspect Ratio Scaling

### Contributors 4



**keyu-wang-2002** Keyu



**huangyangyi** Yangyi Huang



**davidgonmar** David González-Ma...



**Joluck** J.L



**Shiwei Liu**

Faculty@ELLIS Institute Tübingen & MPI for Intelligent Systems  
在 [tue.ellis.eu](mailto:tue.ellis.eu) 的电子邮件经过验证 - [首页](#)  
[Deep Learning](#) [AI Modeling](#) [Low-dimensional Learning](#) [LLMs](#)

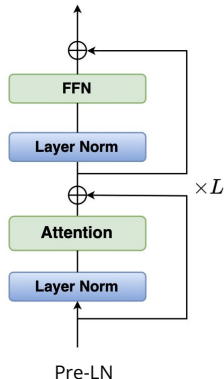
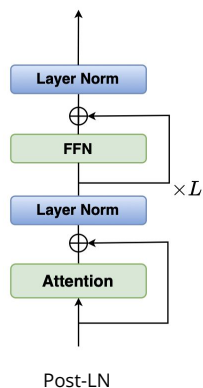


**Weiyang Liu**

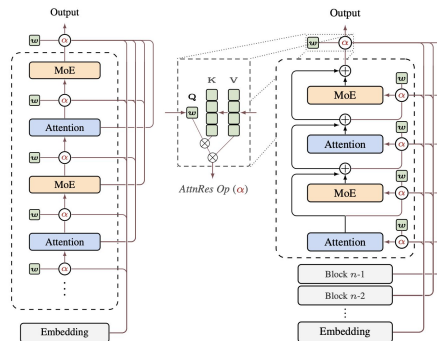
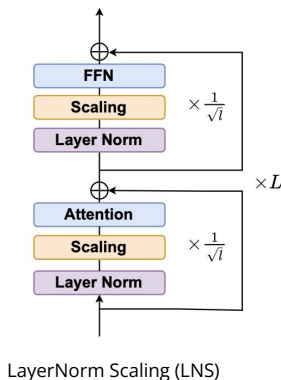
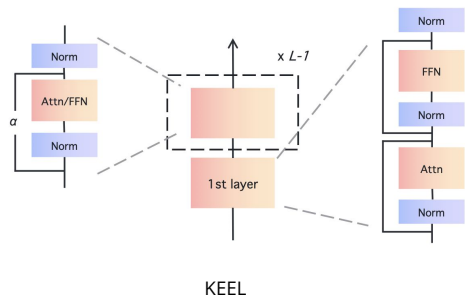
CUHK  
Verified email at [cse.cuhk.edu.hk](mailto:cse.cuhk.edu.hk) - [Homepage](#)  
[Machine Learning](#)

# Project 3: Motivation

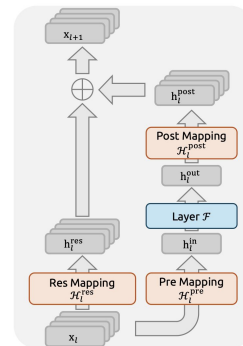
- Which one scales its depth better?



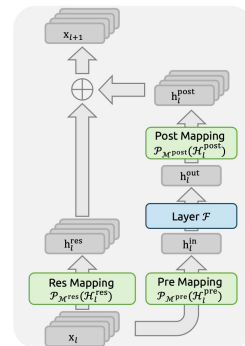
Dominate in modern LLMs



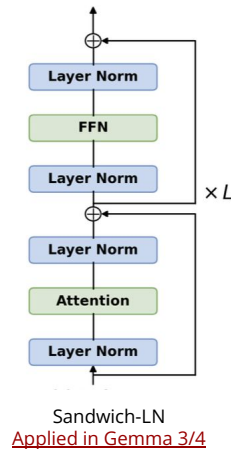
Attention Residuals (AttnRes, left: Full, right: Block)  
Applied in Kimi K3



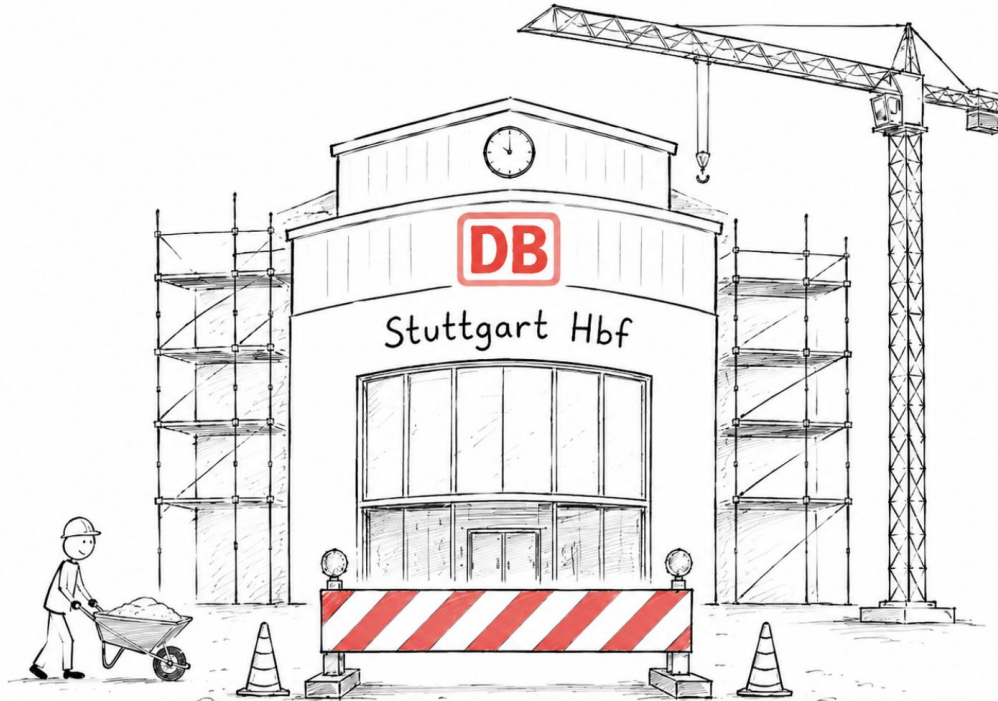
Hyper Connections (HC)  
Applied (Modified) in Owen 3.8



manifold-constrained HC (mHC)  
Applied in Deepseek V4

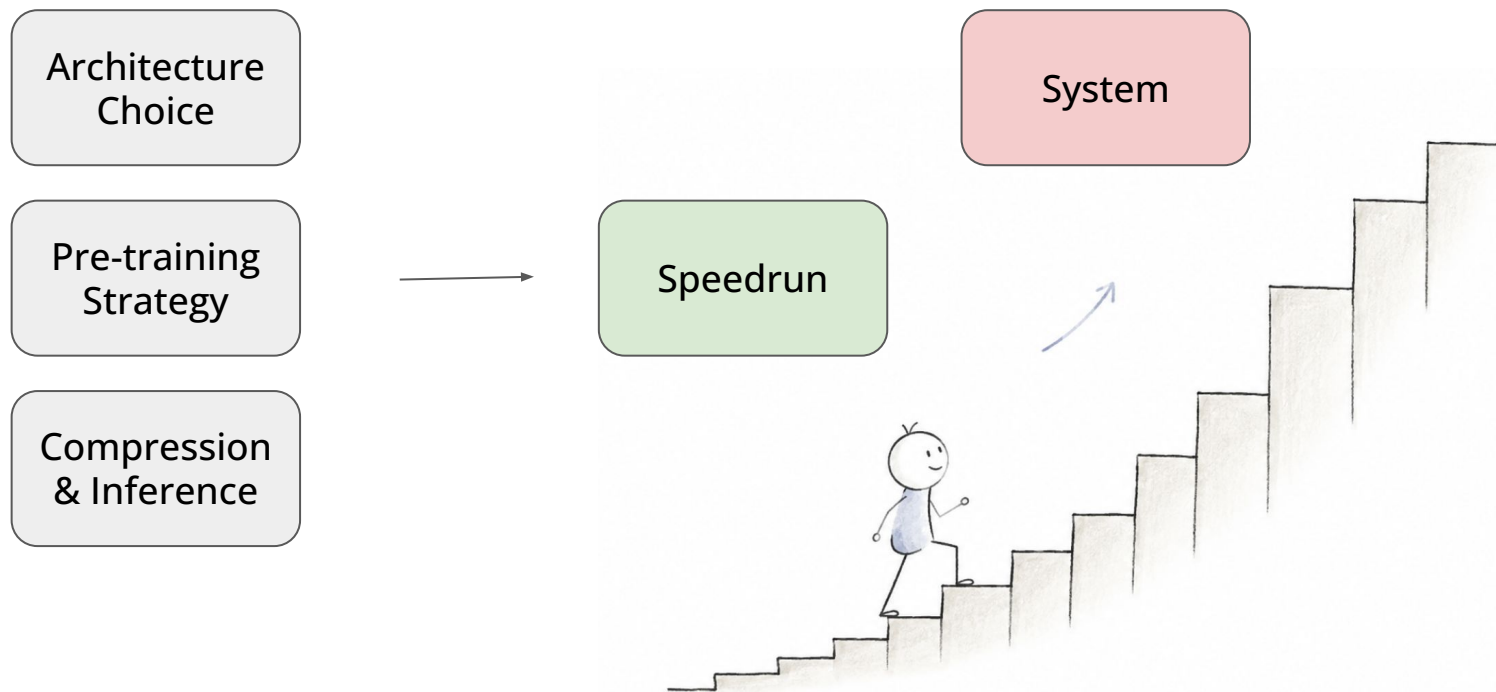


## Project 3: Paper is coming soon!



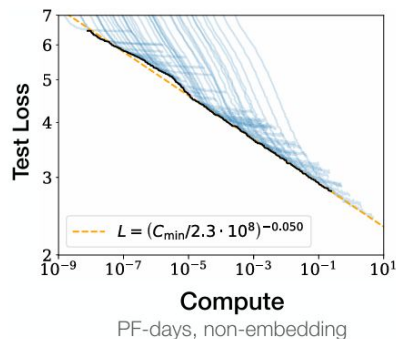
# Towards Effective Depth Scaling in LLMs

2025.06 - 2026.12



# Future Vision

## ➤ Curiosity



observe

explain



predict

control

BTW I'm exploring my master's thesis direction about model dynamics & continual learning.  
Happy to talk.

## ➤ Vision

What we can do to the system that would make the principle reveal whether it is true?  
If we understand a dynamical mechanism well enough, we should eventually be able to control it.

01

Mechanics  
of Learning

What governs the trajectory?

02

Scaling  
of Learning

Which principle is universal?

03

Dynamics  
of Evolution

How does new learning interact  
with existing knowledge?

04

System  
Support

How should learning be  
engineered across the computing?



Thanks :)